Vin Shin

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Education

University of California, Santa Barbara – BS in Electrical Engineering June 2028

- **GPA:** 3.76/4.00 **Awards:** Dean's List

Experience

Low Voltage Electrical Lead – UCSB Gaucho Racing (Formula SAE Electric) June 2025 – Present

- Manage undergraduate electronics team across PCB design, power distribution, driver interface, and sensor integration for FSAE Electric competition vehicle.
- Architect system-level electrical design including enclosures, wire routing, and CAN network topology; reduced connector count 60% (52 to 21) while improving serviceability.
- Coordinate cross-functional development with mechanical, high-voltage, and software teams to meet competition deadlines and technical regulations.

Projects

1.5kW FOC Motor Controller – Personal Project Jan. 2026

- Architecting high-performance FOC inverter for BLDC motors: 30kHz switching at 25A peak AC current from 60V bus using Infineon OptiMOS MOSFETs and UCC27211A gate drivers.
- Developed 4900mm^2 4-layer PCB with 2oz copper inner layers and isolated power/signal planes; cascaded power supply via dual buck converters.
- Implementing STM32G4-based FOC library with synchronized 3-phase ADC sampling, SPI magnetic encoder feedback, and real-time current control.

Battery Management System – UCSB Gaucho Racing Sept. 2025

- Designed schematic and 4-layer PCB for GR26 battery segment modules using NXP MC33771C AFE managing 2p140s configuration; achieved 39% board area reduction versus GR25 design.
- Implemented STM32G474-based thermistor measurement system with 10-sample moving average filtering for $<$0.5°C noise rejection across 14 temperature points per segment.
- Utilized daisy-chain TPL (Transformer Physical Layer) communication across 10 modules for real-time cell monitoring and balancing with capacitive isolation.

GR26 Wire Harness – UCSB Gaucho Racing Dec. 2025

- Developed comprehensive electrical architecture schematic in Rapid Harness & SolidWorks mapping all power, CAN, and analog signal paths across 8 vehicle subsystems.
- Designed wire routing through RapidHarness and SolidWorks Routing, maintaining FSAE-required clearances through chassis while optimizing for serviceability.
- Engineered noise-resistant harness with heat shrink covering, shielded twisted-pair CAN lines, and motorsport-grade Deutsch connectors.

Distributed Sensor Network – UCSB Gaucho Racing Dec. 2024

- Architected CAN-based sensor network for GR25 data acquisition with custom PCB nodes featuring integrated sensor ICs and IP67-rated enclosures.
- Designed modular enclosures using SolidWorks with laser-cut acrylic, silicone gaskets, and threaded inserts enabling field replacement in $<$2 minutes.

Additional Projects: GR26 Attic Enclosure (IP-rated power distribution housing), GR26 Driver Interface PCB (CAN-based dashboard with STM32 & WS2812b LEDs)

Technical Skills

Languages: C/C++, Python, MATLAB, STM32 HAL/LL library
Electrical Design: Altium Designer, Fusion, KiCad, LTspice | 4-layer mixed-signal, high power requirements
Mechanical: SolidWorks, Fusion 360, RapidHarness | Enclosure design, wire harnessing, thermal management
Test Equipment: Oscilloscope, logic analyzer, soldering (SMD/THT), thermal imaging
Technical: Power electronics, Digital Synchronous Systems, Embedded systems (STM32), Signal Processing